# Overview of the BIOMASS R package

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GEO-TREES Panama Workshop

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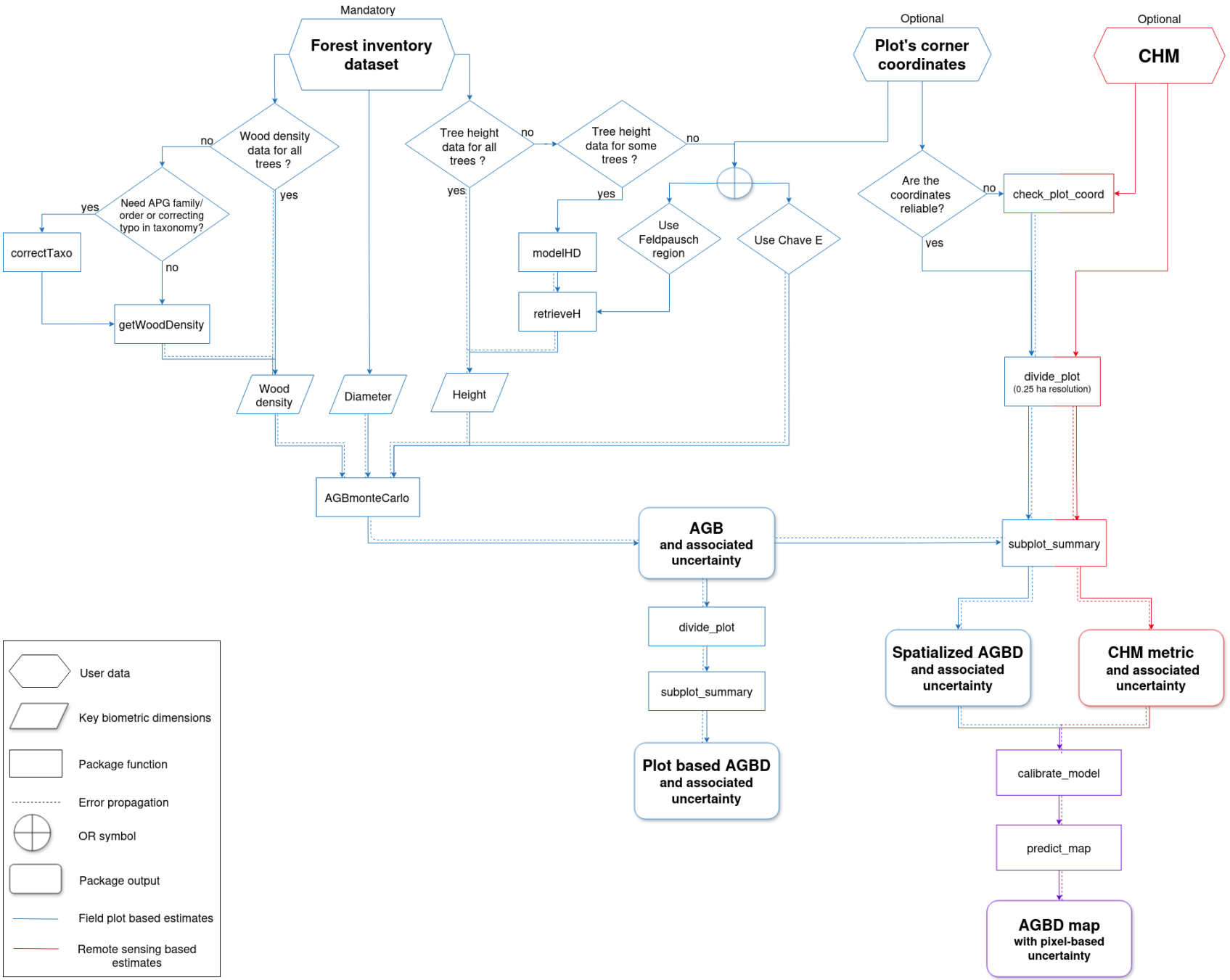
# Brief introduction


Mandatory
Forest inventory dataset
Optional
Plot's corner coordinates
Optional
CHM
Wood density data for all trees ?
no
yes
Need APG family/ order or correcting typo in taxonomy?
yes
no
correctTaxo
getWoodDensity
Tree height data for all trees ?
no
yes
Tree height data for some trees ?
no
yes
modelHD
retrieveH
Use Feldpausch region
Use Chave E
Are the coordinates reliable?
no
yes
check_plot_coord
divide_plot
(0.25 ha resolution)
Wood density
Diameter
Height
AGBmonteCarlo
AGB
and associated uncertainty
subplot_summary
divide_plot
subplot_summary
Plot based AGBD
and associated uncertainty
Spatialized AGBD
and associated uncertainty
CHM metric
and associated uncertainty
calibrate_model
predict_map
AGBD map
with pixel-based uncertainty
User data
Key biometric dimensions
Package function
Error propagation
OR symbol
Package output
Field plot based estimates
Remote sensing based estimates

# CRAN R package

**BIOMASS: Estimating Aboveground Biomass and Its Uncertainty in Tropical Forests**

Contains functions for estimating above-ground biomass/carbon and its uncertainty in tropical forests. These functions allow to (1) retrieve and correct taxonomy, (2) estimate wood density and its uncertainty, (3) build height-diameter models, (4) manage tree and plot coordinates, (5) estimate above-ground biomass/carbon at stand level with associated uncertainty. To cite 'BIOMASS', please use citation('BIOMASS'). For more information, see Réjou-Méchain et al. (2017) <doi:10.1111/2041-210X.12753>.

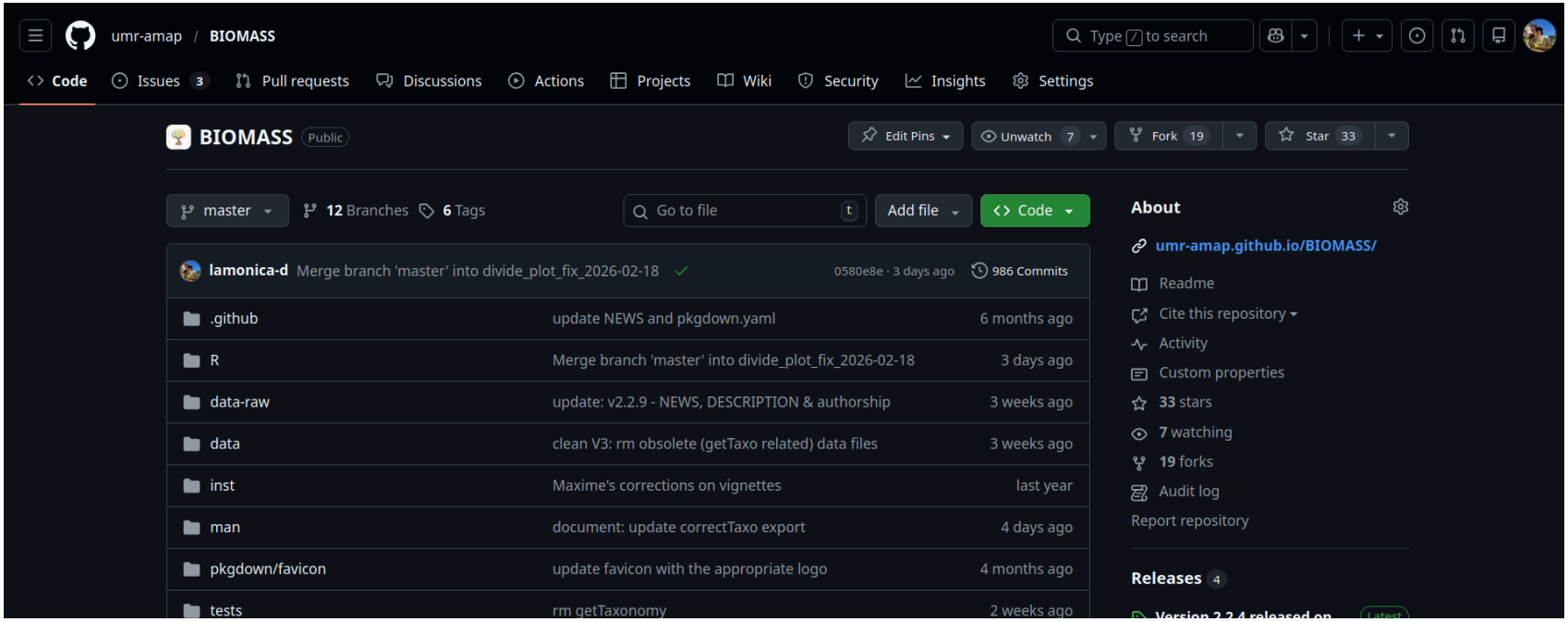
| | |
|---|---|
| Version: | 2.2.4-1 |
| Depends: | R (≥ 3.6) |
| Imports: | minpack.lm, jsonlite, methods, proj4, graphics, stats, utils, data.table (≥ 1.9.8), rappdirs, sf, terra, ggplot2 |
| Suggests: | knitr, rmarkdown, prettydoc, testthat, vdiffr, curl, geodata, httr2, pkgdown, dplyr |
| Published: | 2025-10-10 |
| DOI: | 10.32614/CRAN.package.BIOMASS |
| Author: | Maxime Réjou-Méchain [aut, dtc], Guillaume Cornu [aut], Dominique Lamonica [aut, cre], Arthur Bailly [aut], Arthur Pere [aut], Ariane Tanguy [aut], Jerome Chave [dtc], Ted Feldpausch [dtc], Philippe Verley [ctb], Camille Piponiot [aut], Bruno Hérault [aut] |
| Maintainer: | Dominique Lamonica <dominique.lamonica at ird.fr> |
| BugReports: | https://github.com/umr-amap/BIOMASS/issues/ |
| License: | GPL-2 |
| URL: | https://umr-amap.github.io/BIOMASS/, https://github.com/umr-amap/BIOMASS/ |
| NeedsCompilation: | no |
| Citation: | BIOMASS citation info |
| Materials: | README, NEWS |
| CRAN checks: | BIOMASS results |

**Documentation:**

| | |
|---|---|
| Reference manual: | BIOMASS.html , BIOMASS.pdf |
| Vignettes: | Vignette BIOMASS (source, R code)<br>Spatialize trees and forest stand metrics with BIOMASS (source, R code) |

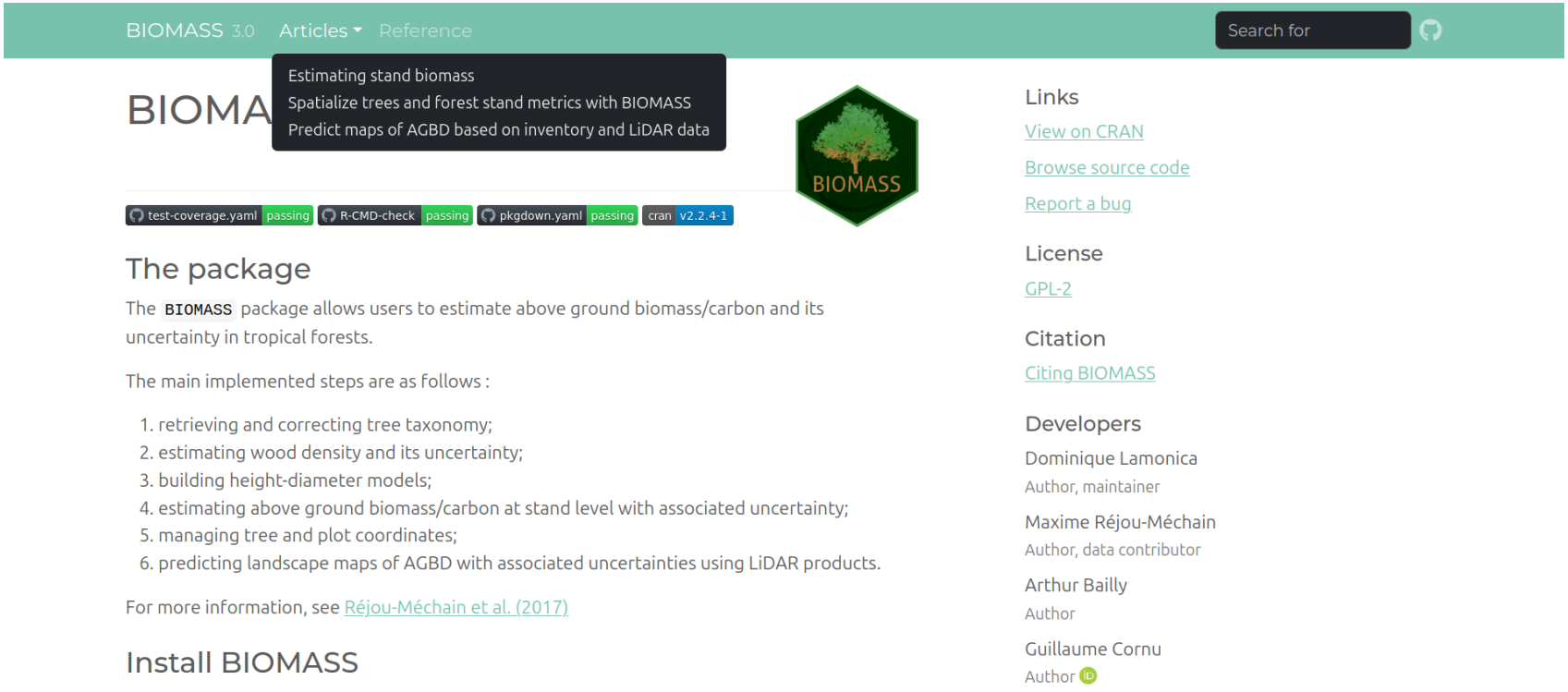
- Many contributors - some on specific features, others on the entire package

# GitHub public repository



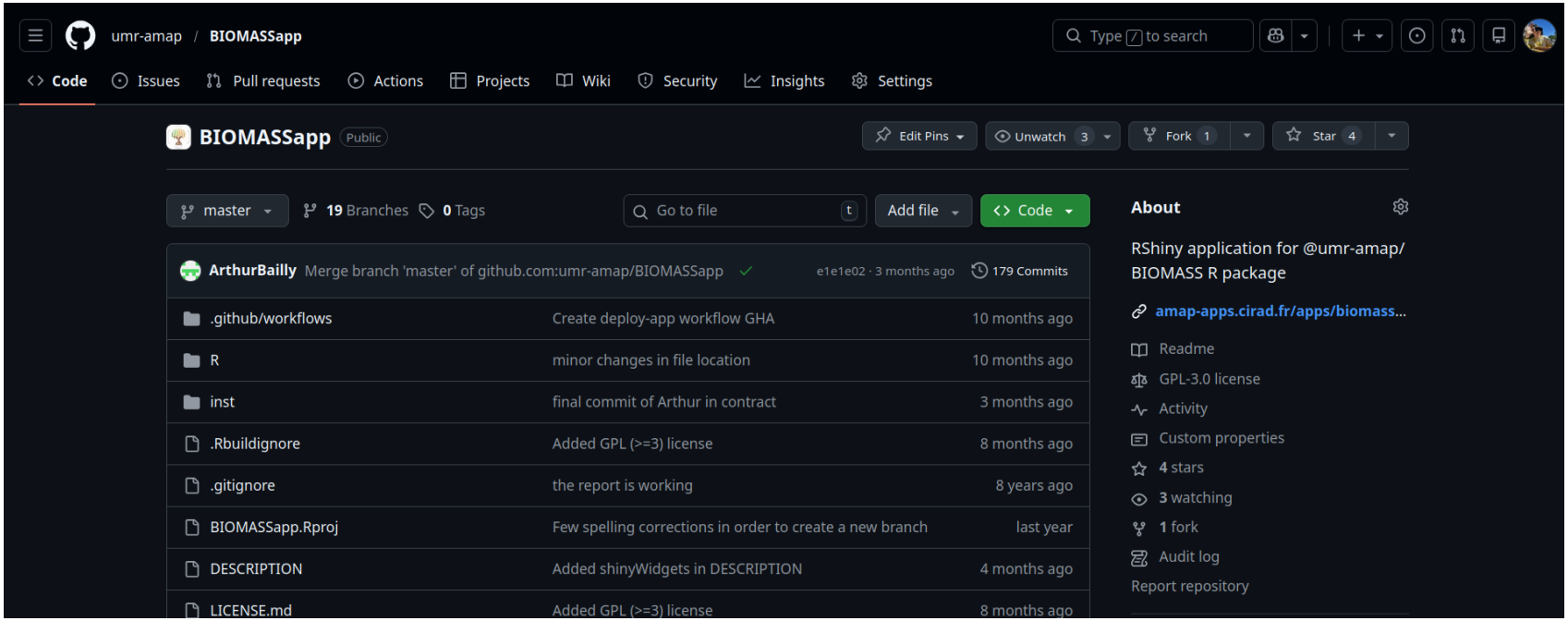

- Entire code accessible, issues open
- Developing version
- CI

# Vignettes

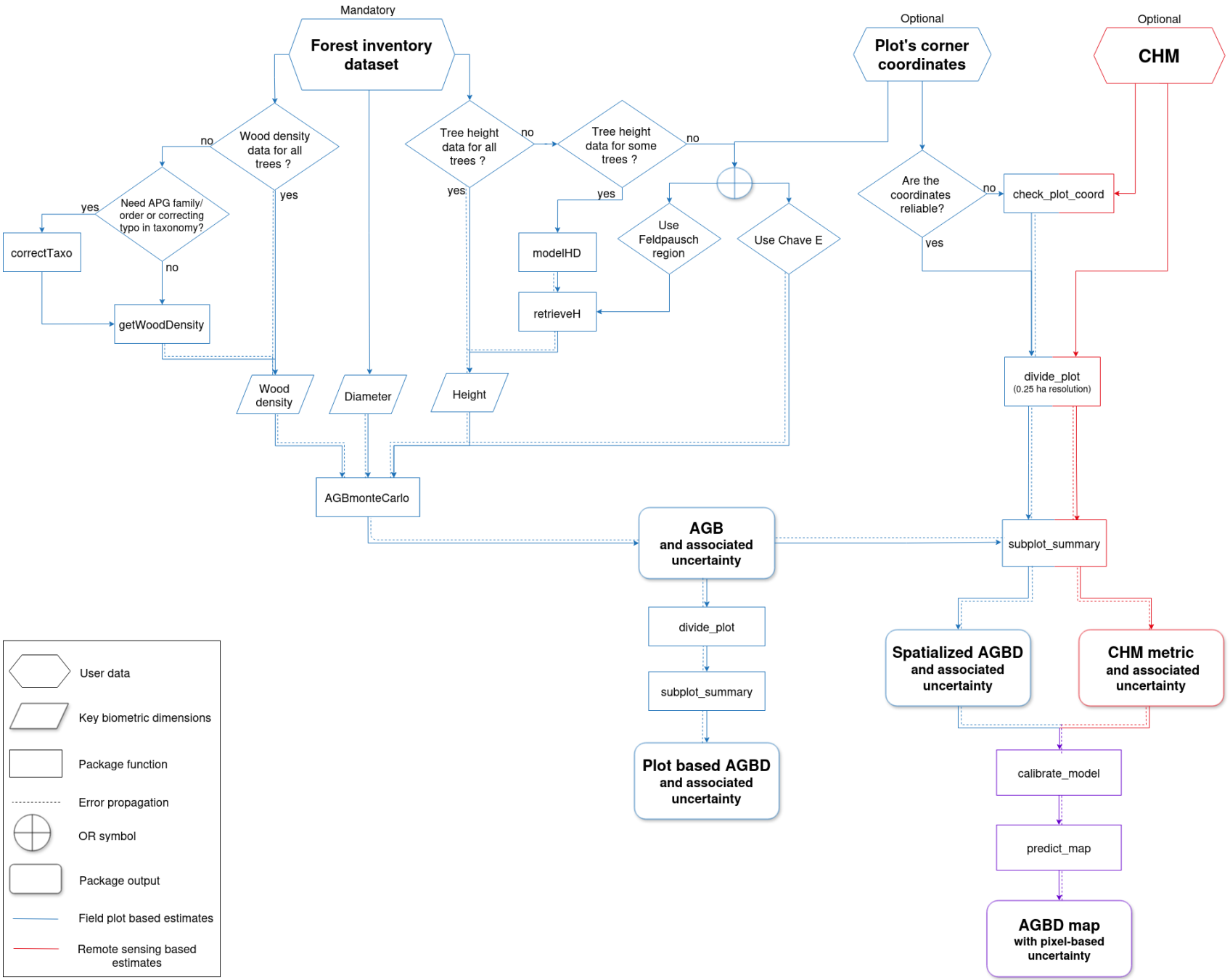



- Tutorial for the entire pipeline divided in three main articles, applied on reduced Nouragues data

# Shiny app - BIOMASSapp




- Estimate AGBD per subplot with uncertainties
- Spatialize stand metrics
- No map prediction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# BIOMASSapp demonstration

# Predict AGBD maps using inventory and LiDAR data

# CHM-AGBD model calibration: proposed statistical framework

- geostatistical model with SPV-I/C (SPatially Varying Intercept/Coefficients) to integrate spatial correlation:
- $y(s) = (\alpha + \tilde{\alpha}(s)) + (\beta + \tilde{\beta}(s)) \times x(s) + \epsilon(s)$
  with $\tilde{\alpha}(s_1), ..., \tilde{\alpha}(s_n) \sim MVN(0, C_\alpha(s_i, s_j))$
- references

Remote Sensing of Environment 169 (2015) 113–127

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

LiDAR based prediction of forest biomass using hierarchical models with spatially varying coefficients

Chad Babcock [a], Andrew O. Finley [b,*], John B. Bradford [c], Randall Kolka [d], Richard Birdsey [e], Michael G. Ryan [f]

Remote Sensing of Environment 318 (2025) 114557

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

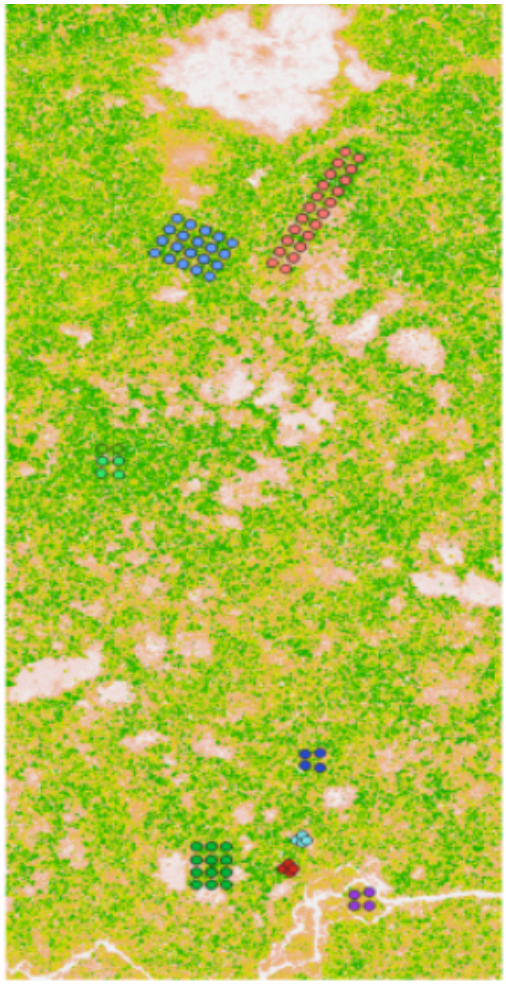
A geostatistical approach to enhancing national forest biomass assessments with Earth Observation to aid climate policy needs

Neha Hunka [a,*], Paul May [b], Chad Babcock [c], José Armando Alanís de la Rosa [d], Maria de los Ángeles Soriano-Luna [d], Rafael Mayorga Saucedo [d], John Armston [a], Maurizio Santoro [e], Daniela Requena Suarez [f], Martin Herold [f], Natalia Málaga [f], Sean P. Healey [g], Robert E. Kennedy [h], Andrew T. Hudak [i], Laura Duncanson [a]

# Implementation in BIOMASS

- inference and predictions done using brms R package
- two functions: 1st calibrates the model with user plot and LiDAR data, 2nd predicts AGBD and its associated uncertainties over the full raster footprint provided by the user
- full continuity of the pipeline: the inputs of those functions are the outputs from the previous steps
- propagate all AGBD uncertainties from the previous steps, within Bayesian framework

# CHM-AGBD model calibration: Nouragues data example



SPV-C model

$log(y(s)) = (\beta + \tilde{\beta}(s)) \times log(x(s)) + \epsilon(s)$

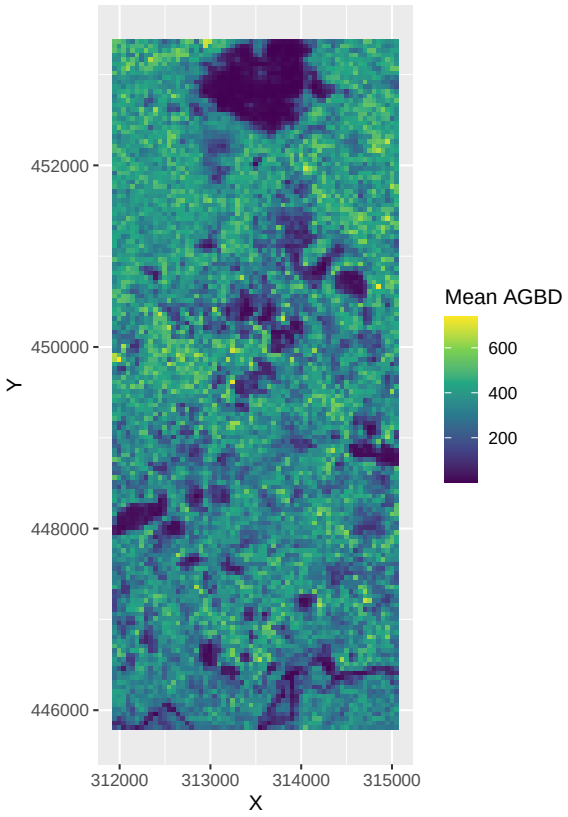
with

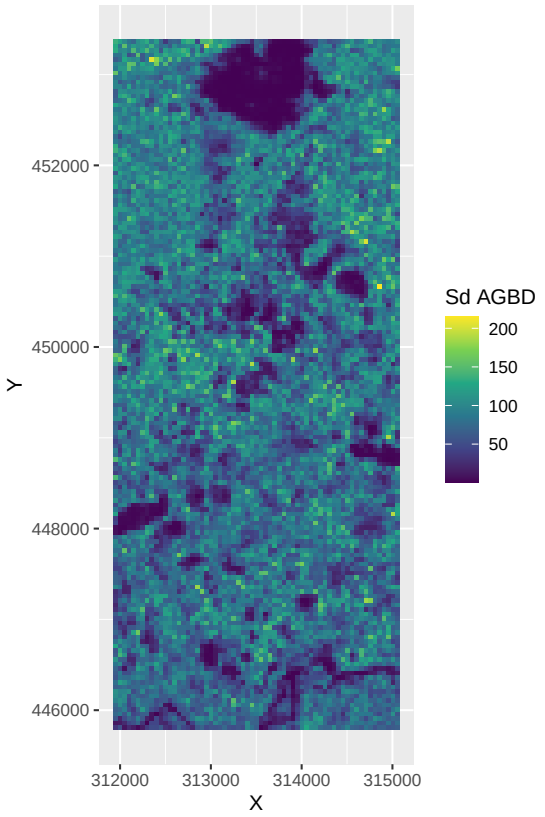
- $y(s)$ AGBD 0.25ha for plot $s$ : ajouter plot de l'agbd carré divisé + incertitudes
- $x(s)$ mean (or median) CHM 0.25ha for plot $s$ : ajouter plot du raster + carré divisé
- $\tilde{\beta}(s_1), ..., \tilde{\beta}(s_n) \sim MVN(0, C_\beta(s_i, s_j))$

# CHM-AGBD model calibration: Nouragues data example

ici screen du model_fit + plot ppc

# Map prediction: Nouragues data example




ici nouveau plot avec mean, sd, coeff var (à faire)

# Validation

In the vignette we propose to calibrate on 70% of data, then validate on 30%, screen de la vignette

Future work, this week and later on (à traduire)

# Future work, this week and later on (à traduire)

- Temporal BIOMASS
- Quality check plus robuste avec du simple ou multidates
- Nouvelle allométrie générique globale de l'AGB
- Implementation d'une (ou de) nouvelle(s) approche(s) d'estimation de hauteur en absence de données terrain
- Travail sur le modèle AGBD - LiDAR déjà développé dans BIOMASS V3.0 (flexibilité)
- Prise en compte l'incertitude de positionnement sur individual trees
- Gestion de l'API WFO pour la correction taxonomique
- Gestion de plots circulaires dans le pipeline de spatialisation des estimations
- Travail de recherche autour de la mise en place d'une nouvelle stratégie de modélisation AGB & LiDAR